

Extension 1

MATHEMATICS

YEAR 11

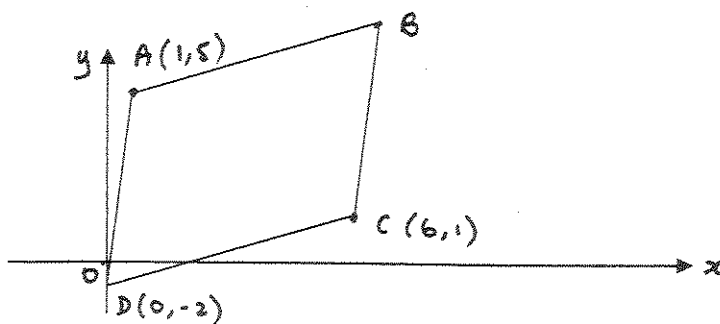
PRELIMINARY ASSESSMENT TASK 2

July 2005

Question 1

10 marks

- a. Give the exact value of 4
- i. $\cot 300^\circ$
 - ii. $2\sin 15^\circ \cos 15^\circ$
- b. Differentiate $\frac{3x^2 + x}{x}$ 1
- c. Evaluate $\lim_{x \rightarrow \infty} \frac{4x^3 + 2x^2 - 1}{3 - 2x^2 + x^3}$ 1
- d. ABCD is a parallelogram 2
- i. Find the midpoint of diagonal AC
 - ii. Find the coordinates of the point B



- e. Let A be the point $(-1, 2)$ and B be the point $(4, 9)$. Find the coordinates of the point P which divides the interval AB internally in the ratio $3 : 2$ 2

Question 2

10 marks

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- a. Find the acute angle, α , between the two lines, $2x - y + 4 = 0$ and $y = 1 - x$. Give your answer correct to the nearest minute. 3
- b. Find the exact value of $\sin 105^\circ$ 2
- c. Find $\frac{d}{dt}(at^3 - 2a + t)$ 1

d. If $y = (x^2 + 1)^3$, find $\frac{dy}{dx}$ 2

e. Differentiate $f(x) = \frac{x^2}{1-x^2}$ 2

Question 3

10 marks

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a. Find the gradient of the normal to the curve $y = x\sqrt{x}$ at the point (4, 8) 2

b. Find the equation of the tangent to the curve $y = x^2 + \frac{5}{x} - 2$ at the point P (1, 4) 2

c. Given that $0 < x < 45^\circ$, prove that 3

$$\tan(x + 45^\circ) = \frac{\cos x + \sin x}{\cos x - \sin x}$$

d. Find all angles θ , with $0^\circ \leq \theta \leq 360^\circ$, which $\sin 2\theta = \sin \theta$. 3

Question 4

10 marks

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a. 4

i. Differentiate $f(x) = \sqrt{x}$ by first principles

$$\text{given } \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

ii. For what values of x is $\frac{d}{dx}(\sqrt{x})$ undefined?

b. 4

i. Show that $(\sin x - \cos x)^2 = 1 - \sin 2x$

ii. Hence, or otherwise, find the value of $\sin 15^\circ - \cos 15^\circ$, in simplest exact form.

c. If $\cos x = \frac{3}{4}$ and $\sin x < 0$, find the exact value of $\sin 2x$ 2

Question 5

10 marks

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a.

5

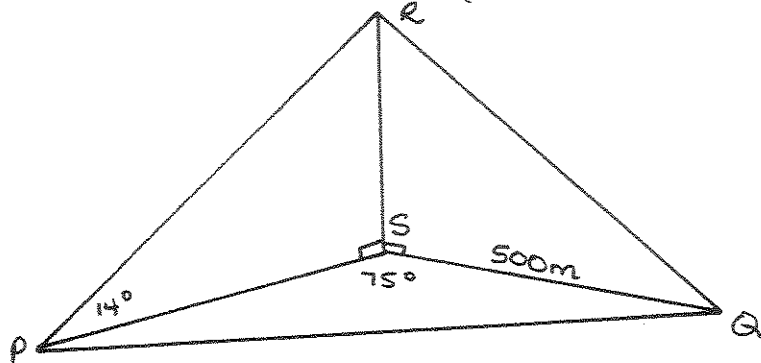
- i. Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$ if $R > 0$ and $0 < \alpha < 90^\circ$
- ii. Hence, or otherwise, solve $\cos x - \sqrt{3} \sin x = 2$, for $0^\circ \leq x \leq 360^\circ$

b.

5

Two houses, P and Q, lie in the same plane as S, the foot of a tower RS. The angle of elevation from P to the top of the tower is 14° and the angle of depression from the tower to Q is 18° .

It is known that the distance from S to Q is 500 metres.



- i. Find an expression for the height of the tower.
- ii. Find the distance from P to S correct to the nearest centimetre.
- iii. Given that P and Q subtend an angle of 75° at S, how far apart are the two houses, correct to the nearest metre?

Question 6

10 marks

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a.

5

- i. If $\tan \frac{\theta}{2} = t$, state the results for $\sin \theta$ and $\cos \theta$
- ii. Using these results show that $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$
- iii. Hence, find the exact value of $\tan 15^\circ$

b.

5

- i. Show that the perpendicular distance from the point $(4, 5)$ to the line $y = mx$ is given by,

$$d = \frac{|4m - 5|}{\sqrt{m^2 + 1}}$$

- ii. The line $y = mx$ is a tangent to the circle $(x - 4)^2 + (y - 5)^2 = 4$, explain why $\frac{|4m - 5|}{\sqrt{m^2 + 1}} = 2$

- iii. Hence, show that m satisfies the equation $12m^2 - 40m + 21 = 0$

Question 1

a. i. $\cot 300^\circ = \frac{1}{-\tan 60^\circ}$
 $= -\frac{1}{\sqrt{3}}$ ✓

ii. $2\sin 15^\circ \cos 15^\circ = \sin 2(15^\circ)$
 $= \sin 30^\circ$
 $= \frac{1}{2}$ ✓

b. $d/dx(3x+1) = 3$ ✓

c. $\text{Lim} = 4$ ✓

d. i. $(7/2, 3)$ ✓

ii. $\frac{0+x}{2} = \frac{7}{2}$ $\frac{-2+y}{2} = 3$

$x = 7$ $-2+y = 6$
 $y = 8$

$B(7, 8)$ ✓

2. $A(-1, 2)$ $B(4, 9)$
 $3:2$

P is $\left(\frac{-1 \times 2 + 4 \times 3}{5}, \frac{2 \times 2 + 9 \times 3}{5} \right)$
 $= (2, 3\frac{1}{5})$

Question 2

a) $y = 2x + 4$ $y = 1 - x$ ✓
 $m_1 = 2$ $m_2 = -1$

$\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ ✓

$= \left| \frac{2 - (-1)}{1 + 2 \times (-1)} \right|$

$= \left| \frac{3}{-1} \right|$

$\tan \alpha = 3$

$\alpha = 71^\circ 34'$ ✓

b) $\sin 105^\circ = \sin(45 + 60)$ ✓
 $= \sin 45 \cos 60 + \cos 45 \sin 60$
 $= \frac{1}{\sqrt{2}} \times \frac{1}{2} + \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}$
 $= \frac{1 + \sqrt{3}}{2\sqrt{2}}$ ✓
 $= \frac{\sqrt{2} + \sqrt{6}}{4}$

c) $d/dt(at^3 - 2a + t) = 3at^2 + 1$ ✓

d) $\frac{dy}{dx} = 5(x^2 + 1)^4 \cdot 2x$ ✓
 $= 10x(x^2 + 1)^4$

e) $u = x^2$ $u' = 2x$ ✓
 $v = 1 - x^2$ $v' = -2x$

$f'(x) = \frac{(1-x^2) \cdot 2x - x^2 \cdot (-2x)}{(1-x^2)^2}$
 $= \frac{2x - 2x^3 + 2x^3}{(1-x^2)^2}$
 $= \frac{2x}{(1-x^2)^2}$ ✓

Question 3

a. $y = x^{3/2}$
 $\frac{dy}{dx} = \frac{3}{2} x^{1/2}$
 $= \frac{3\sqrt{x}}{2}$ ✓ at $x=4$

$M_{\text{tangent}} = \frac{3 \cdot \sqrt{4}}{2}$
 $= 3$

∴ $M_N = -1/3$ ✓

b. $y' = 2x - \frac{5}{x^2}$ at $x=1$

$M_T = 2 \times 1 - \frac{5}{1}$
 $= 2 - 5$
 $= -3$ ✓

∴ $y - 4 = -3(x - 1)$ ✓
 $y = -3x + 7$

∴ LHS = $\tan(x + 45)$
 $= \frac{\tan x + \tan 45}{1 - \tan x \cdot \tan 45}$ ✓
 $= \frac{\tan x + 1}{1 - \tan x \cdot 1}$
 $= \frac{\sin x / \cos x + 1}{1 - \sin x / \cos x}$ ✓ $\left(\frac{\cos x}{\cos x} \right)$
 $= \frac{\sin x + \cos x}{\cos x - \sin x}$
 $= \frac{\cos x + \sin x}{\cos x - \sin x}$ ✓
 $= \text{RHS}$

d) $\sin 2\theta = \sin \theta$

$2\sin \theta \cos \theta - \sin \theta = 0$

$\sin \theta (2\cos \theta - 1) = 0$ ✓

$\sin \theta = 0$ $\cos \theta = 1/2$

$\theta = 0, 180^\circ, 360^\circ$ ✓ $\theta = 60^\circ, 300^\circ$ ✓

∴ $\theta = 0, 60^\circ, 180^\circ, 300^\circ, 360^\circ$

Question 4

a. $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$ ✓
 $= \lim_{h \rightarrow 0} \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})}$ ✓
 $= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$
 $= \frac{1}{\sqrt{x} + \sqrt{x}}$
 $= \frac{1}{2\sqrt{x}}$ ✓

ii. undefined when $x \leq 0$ ✓

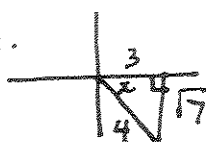
b.

i. LHS = $(\sin x - \cos x)^2$
 $= \sin^2 x - 2\sin x \cos x + \cos^2 x$ ✓
 $= 1 - 2\sin x \cos x$
 $= 1 - \sin 2x$ ✓

ii. using result in part (i)

$(\sin 15^\circ - \cos 15^\circ)^2 = 1 - \sin 30^\circ$
 $\sin 15^\circ - \cos 15^\circ = \pm \sqrt{1 - 1/2}$
 $= \pm \sqrt{1/2}$

take neg $= -1/\sqrt{2}$ ✓
 as $\cos 15^\circ > \sin 15^\circ$ (one for the neg)

c.  4th quad ✓

$\sin 2x = 2\sin x \cos x$
 $= 2 \times -\frac{\sqrt{7}}{4} \times \frac{3}{4} = -\frac{3\sqrt{7}}{8}$ ✓

a. i. $R = \sqrt{1 + \sqrt{3}^2} = 2 \checkmark$

$\tan \alpha = \sqrt{3}$

$\alpha = 60^\circ \checkmark$

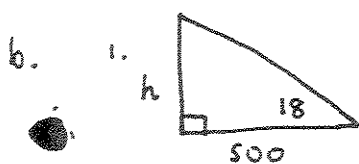
$\therefore 2 \cos(x + 60^\circ)$

ii. $2 \cos(x + 60^\circ) = 2 \checkmark$

$\cos(x + 60^\circ) = 1$

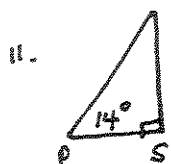
$x + 60^\circ = 0, 360^\circ \checkmark$

$\therefore x = 300^\circ \checkmark$



$\frac{h}{500} = \tan 18^\circ$

$h = 500 \tan 18^\circ \checkmark$



$\tan 14^\circ = \frac{h}{PS} \checkmark$

$PS = h \div \tan 14$

$= \frac{500 \tan 18^\circ}{\tan 14^\circ} \checkmark$

$= 651.59 \text{ m}$



$PQ^2 = 651.59^2 + 500^2 - 2 \times 651.59 \times 500 \times \cos 75^\circ \checkmark$

$PQ^2 = 505925.62 \dots$

$PQ = 711.284 \dots$

$= 711 \text{ m} \checkmark$

Question 6

a. $\sin \theta = \frac{2t}{t^2 + 1} \checkmark$

$\cos \theta = \frac{1 - t^2}{1 + t^2} \checkmark$

$\frac{\sin \theta}{\cos \theta} = \frac{1 - \left(\frac{1 - t^2}{1 + t^2} \right)}{\frac{2t}{1 + t^2}} \checkmark$

$= \frac{1 + t^2 - (1 - t^2)}{2t} \checkmark$

$= \frac{1 + t^2 - 1 + t^2}{2t}$

$= \frac{2t^2}{2t}$

$= t$

$= \tan \theta/2$

$= \text{RHS.}$

iii. $\tan 15^\circ = \tan (30/2)$

$= \frac{1 - \cos 30}{\sin 30}$

$= \left(1 - \frac{\sqrt{3}}{2} \right) \times \frac{2}{1}$

$= 2 - \sqrt{3} \checkmark$

b. $mx - y = 0$

i. $d_{\perp} = \left| \frac{m \cdot 4 - 5 + 0}{\sqrt{m^2 + (-1)^2}} \right| \checkmark$

$= \frac{|4m - 5|}{\sqrt{m^2 + 1}}$

ii. $C = (4, 5) \quad r = 2$

$\therefore$ perp. dist from centre to circumf = radius.

$\therefore \frac{|4m - 5|}{\sqrt{m^2 + 1}} = 2$

iii. $\frac{|4m - 5|}{\sqrt{m^2 + 1}} = 2$

$\frac{\sqrt{(4m - 5)^2}}{\sqrt{m^2 + 1}} = 2$

$(4m - 5)^2 = 4(m^2 + 1)$

$16m^2 - 40m + 25 = 4m^2 + 4$

$\therefore 12m^2 - 40m + 21 = 0$